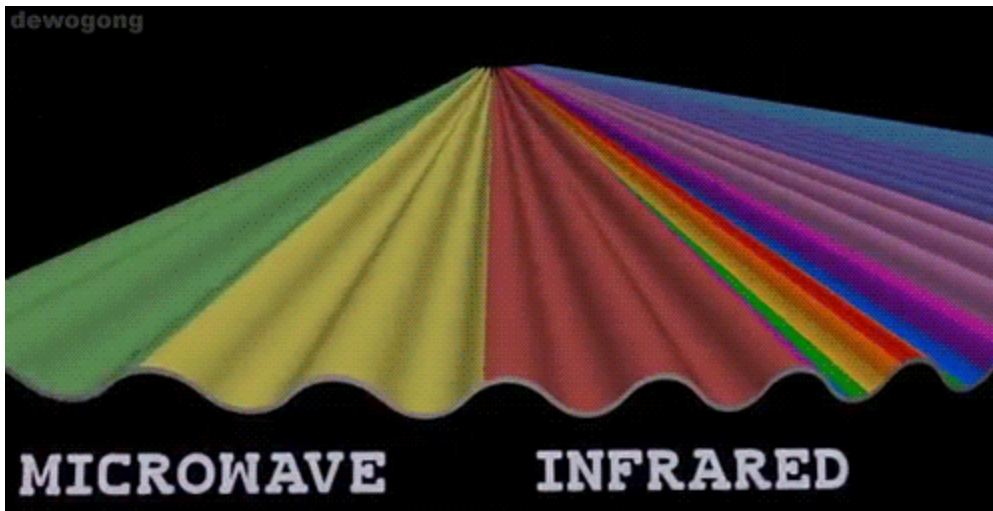


farewell assignment electromagnetic spectrum A↓

⚠ This is a preview of the published version of the quiz

Started: May 3 at 12:26pm

Quiz Instructions



Question 1 10 pts

2) Electromagnetic waves consist of

- A) oscillating electric and magnetic fields.
- B) particles of light energy.
- C) high-frequency sound waves.
- D) compressions and rarefactions of electromagnetic pulses.



D



C



B



A



Question 2 10 pts

3) A changing electric field induces a changing

A) magnetic field.

C) both of these

B) electromagnetic field.

D) neither of these

☐

C

☐

D

☐

B

☐

A



Question 3 10 pts

4) In free space, electromagnetic waves travel at a

A) speed depending on frequency.

B) variety of speeds.

C) single speed.

☐

D

☐

C

☐

B

☐

A



Question 4 10 pts

5) Which of the following is fundamentally different from the others?

A) X-rays

B) gamma rays

C) sound waves

D) radio waves

E) light waves

☐

A

☐

B

☐

C

☐

D

☐

Question 5 10 pts

6) Which of these electromagnetic waves have the shortest wavelength?

A) X-rays

B) infrared waves

C) light waves

D) ultraviolet waves

E) radio waves

☐

C

☐

D

☐

A

☐

B

☐

Question 6 10 pts

7) Which of these electromagnetic waves have the longest wavelength?

A) X-rays

B) radio waves

C) light waves

D) infrared waves

E) ultraviolet waves

☐

B

☐

D

☐

C



A



Question 7 10 pts

Consider a light source. Let's say an IR lamp used for reptiles or for therapy. The Power is 40W. The energy will flow in space in 3D. The same energy produced by the source will spread out over a sphere of radius R and area $4\pi R^2$. The energy flowing through a unit area at a distance R is therefore $P / (4\pi R^2)$

So at a distance $R=50\text{cm}$. The energy flowing through a unit area is:

(remember for final)



about 13W/m^2



about 8W/m^2



about 40W/m^2

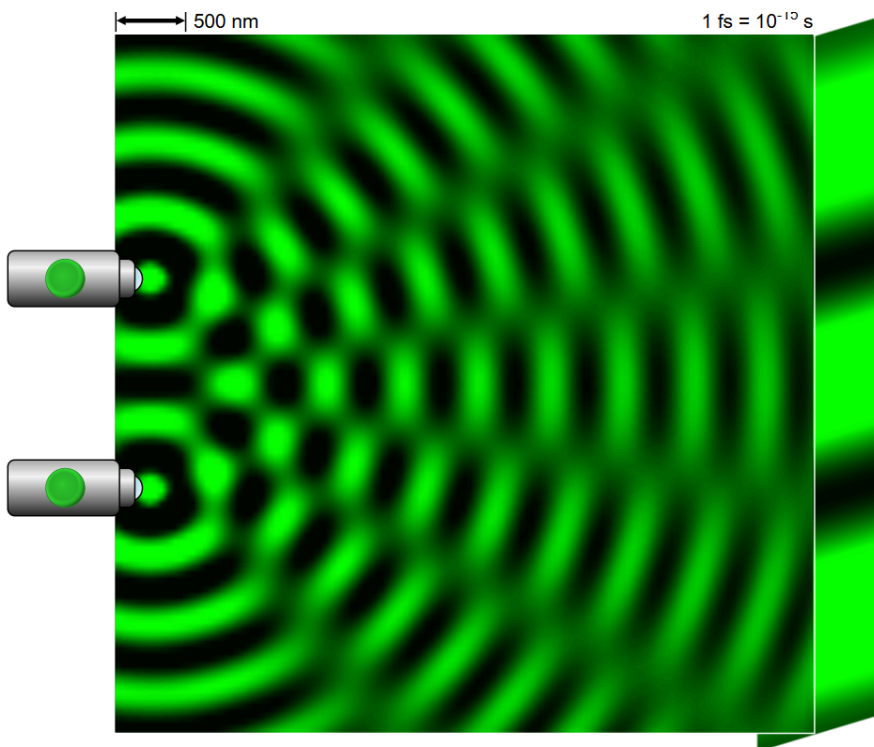


about 126W/m^2



Question 8 10 pts

That experiment (done with light.)shows which property of waves?





reflection



polarization



interference



refraction



Question 9 10 pts

$$\oint \vec{B} \cdot d\vec{a} = 0$$

This equation shows that



a magnetic field circulates around a current



a magnetic field always diverge or converge



the circulation of a static magnetic field is 0



there is no monopole



Question 10 10 pts

What do you find when you compute:

$$\frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

See constants in equation sheet.



speed of light



speed of light squared



the frequency of FM radio



a random number



Question 11 5 pts

Four Maxwell equations in space (no current) are shown on a dark blue background with yellow boxes for the equations and handwritten symbols next to them:

- Top Left: $\int \vec{E} \cdot d\vec{A} = 0$ with a handwritten symbol ∇ .
- Top Right: $\int \vec{B} \cdot d\vec{A} = 0$ with a handwritten symbol $\subset$.
- Bottom Left: $\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$ with a handwritten symbol $\mathbb{B}$.
- Bottom Right: $\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$ with a handwritten symbol $\mathcal{P}$.

Here are the very elegant Maxwell equations in space (no current)

Which one says that an magnetic field circulates around a changing electric field?



A



C



D



B

Not saved

Submit Quiz